CSCI407 Final Project Report

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CSCI4070 Intro to AI

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May 6, 2026

Introduction

This project is a web-based application that uses machine learning to predict fitness-related outcomes such as BMI and weight category. The system works by allowing the user to upload a CSV file, then select which variables will be used as predictors and which one will be the target variable. Based on this selection, the user can choose between different algorithms, including linear regression, logistic regression, and decision tree models, to train the model and generate predictions.

The application also includes a user input feature where users can enter their own values, such as height in centimeters, number of workout days per week, and sleep hours. These inputs are used by the trained model to produce a prediction. This makes the system more interactive and allows users to see how different inputs affect the predicted results.

Problem Statement

Maintaining a healthy body weight is a common-world problem because weight change is influenced by several lifestyle factors, including calories, exercise habits, sleep, water intake, and more. Many individuals can see their current health data, but they may not understand how these factors relate to future weight or BMI categorization, and how to optimize their lifestyle to meet their personal goals. The aim of this project is to create a web application that helps users explore fitness data and make machine learning-based predictions to guide both general information as well as personalized applications.

Solution Approach

The system works by allowing the user to either upload a dataset in CSV format or enter their own input values. After uploading the data, the user selects the predictor variables and the target the target variable that the model will use. The user then chooses a machine learning algorithm, such as linear regression, logistic regression, or a decision tree model, depending on the type of prediction needed. Once these selections are made, the system trains the model and generates a prediction. If the user chooses to enter their own data, values such as height, workout days per week, and sleep hours used to produce a prediction based on the trained model.

Dataset Description

The CSV file contains users' basic information about their routine, such as their height, weight, age, sleep hours, how many days per week they exercise, and how many minutes they work out each day. It also includes part of the user's diet, such as how many liters of water and how many grams of protein they consume.

The dataset includes weight and BMI information. Weight is calculated in kilograms, and the BMI is calculated using the height and weight. The BMI makes it possible to calculate a weight category. The dataset gives the system information about the user's lifestyle and physical health. By combining routine, diet, and body measurements, the model can relate these values to create an expected final weight and weight category.

Data Preprocessing

Data preprocessing is a crucial step in preparing the dataset for machine learning models. In this project, several preprocessing techniques were applied to ensure data quality and improve model performance.

First, missing values in the dataset were handled appropriately. Any incomplete or null entries were either removed or replaced using suitable methods such as mean or median imputation, depending on the type of variable. This step ensured that the model training process was not affected by incomplete data.

Next, label encoding was applied to convert categorical variables into numerical format. Since machine learning algorithms cannot process text-based categories directly, categorical values such as classification labels were transformed into numeric representations. This allowed algorithms like Logistic Regression and Decision Trees to interpret the data correctly.

Finally, the dataset was divided into training and testing subsets using a 70/30 split. Specifically, 70% of the data was used to train the model, while the remaining 30% was reserved for testing and evaluation. This approach helps assess how well the model generalizes to unseen data and prevents overfitting.

Machine Learning Models

The application gives users the option to select one of three algorithms: linear regression, logistic regression, and decision tree. Linear Regression is favorable when the target variable is numeric, such as final_weight_kg or bmi. Linear Regression is appropriate for estimating a continuous variable based on multiple predictor variables. Logistic Regression is used for classification, in reference to the dataset the categorical variable is weight_category. The project improves Logistic Regression performance by using scaling, feature selection, and a stronger relationship between BMI and the weight_category target. Decision Tree is included as another supervised model option that works well with numerical predictors in this application.

The model training process uses a 70/30 split. The train_test_split function from scikit-learn divides arrays or matrices into random training and testing subsets. In this project, 70% of the dataset provided by the user is used to train the selected model, while 30% is held for evaluation. This follows the assignment requirement and helps estimate how well the model performs on unseen data (in this scenario this would be the individual user input, not the CSV file) instead of only measuring memorization of the training data.

System Design (Architecture)

The application follows a client-server architecture. The frontend is built using HTML and CSS, which allows users to interact with the system by uploading a dataset, selecting variables, and entering input values, while also providing a clean and organized layout. The backend is developed using Python with Flask, which handles user requests, processes the data, and runs the selected machine learning model. The overall flow of the system starts with the user sending data to the server, where the model is trained and used to generate predictions. The results are then sent back to the frontend and displayed to the user.

Results Analysis

To evaluate the performance of the machine learning models, multiple metrics were used, including accuracy, precision, recall, mean squared error (MSE), and coefficient of determination (R^2). These metrics allow comparison between models for both classification and regression tasks.

For regression tasks, Linear Regression performed strongly in predicting numeric outcomes such as final weight. The model achieved an R^2 value of approximately 0.97 and a mean squared error (MSE) of about 7.71, indicating a very strong fit between predicted and actual values. Similarly, Decision Tree Regression also performed well, achieving an R^2 value of approximately 0.97 for BMI prediction.

For classification tasks, Logistic Regression showed high performance in predicting weight categories. After applying feature selection and scaling, the model achieved approximately 0.91 accuracy, 0.91 weighted precision, and 0.91 weighted recall. In comparison, Decision Tree Classification performed lower, with an accuracy of approximately 0.83 on the same dataset.

These results highlight the importance of selecting appropriate models based on the problem type. Linear Regression is more suitable for continuous numeric predictions, while Logistic Regression performs better for categorical classification tasks such as predicting weight categories (e.g., Underweight, Healthy, Overweight, Obese). Although Decision Trees provide interpretability through rule-based decision making, they may not always achieve the highest accuracy depending on the dataset characteristics.

Overall, the project demonstrates model comparison effectively through the user interface by displaying performance metrics and example predictions, allowing users to understand the strengths and limitations of each algorithm.

Discussion

The application is very user-friendly and interactive, making it easy to use. The user can upload different CSV files, select predictor variables from optimal suggestions, and pick a machine learning algorithm. This makes it possible to test and compare how different models perform.

One limitation of the project is that the results depend on the size and quality of the dataset. Depending on the dataset used, the results may not perform well on new data. If the models were tested with more datasets and different settings, the results would be even more accurate.

Conclusion

The Fitness and Weight Prediction Web Application successfully demonstrates a machine learning solution to a real-world health problem. The system allows users to upload data, select algorithms, train models using a 70/30 split, and generate predictions with performance metrics and visualizations.

Linear Regression performed best for numeric predictions such as weight and BMI, while Logistic Regression showed strong results for classification tasks like weight category prediction. Decision Trees provided interpretable results but achieved lower accuracy in comparison.

Overall, the project effectively highlights the importance of choosing the right model for different types of problems. A minor improvement would be adding JavaScript to fully meet front-end requirements.